

Figure 3: Component viewer (Screen1)

If you have dragged and placed everything, the layout will look something like what's shown in Figures 1 and 2.

Make the necessary property changes like we did when changing the text properties for label and button components in previous articles in this series.

Renaming the components helps to identify them in the blocks editor.

So now, your graphical user interface is ready. Figure 1 shows exactly how the application will look after the installation.

The hierarchy of the components that we have dragged to the designer is illustrated in Figures 3 and 4.

After having read the previous articles on App Inventor, I am sure you are pretty clear about how to prepare a GUI. It isn't a complex task — the components just have to be placed on the viewer and a few property changes are required to make them look clearer.

So let's move on and add these behaviours using the blocks editor. I hope you remember how to switch from designer to the blocks editor. There is a button available right above the *Properties* pane to do this.

But before we move on, I would

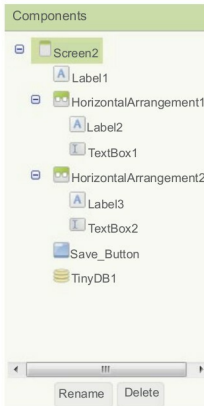


Figure 4: Component viewer (Screen2)

like to go over what we want from our application. Listed below are the functionalities that we are looking for:

- On pressing the button, a custom SMS should be sent to the number already saved in the database.
- On pressing the same button, a phone call should be placed to another number saved in the database.
- By any means, the user should be able to change the emergency contact numbers.
- If the people whose numbers have been saved for emergencies are not available, the user should be duly notified.

Blocks in the blocks editor

I have already prepared the blocks for you. All you need to do is to drag the relevant blocks from the left side palette and drop them on the viewer. Arrange the blocks exactly as shown in the images (Figures 5, 6, 7 and 8). I will explain what each does and how it is called.

The blocks in these figures may look a bit daunting; so let me help you

understand them.

Screen1: When the *Help* button is clicked, the following things will happen (depicted in Figures 5 and 6).

It will generate a custom text message for the texting component, taking the current latitude and longitude values from the location sensor.

Next, it will decide which number the text message should be sent to, and will then fetch the phone number from the database.

Similarly, for the call component, we will need a phone number to which the call should be placed.

It is a good practice to think of all the user scenarios and be prepared in advance; so if such a scenario occurs, the user can be notified to take the necessary action.

The next lines will check whether the fields returned by the DB component are empty. If they are, then the user should be notified to take appropriate action.

If the phone number fields are not empty, we should proceed with the text message and phone call facility.

When the *Help* button is given a long press, it will open Screen2 where the user has set the emergency numbers.

Don't forget to create Screen2 in advance though.

Screen2: When the *Save* button is clicked, the following things will happen (depicted in Figures 7 and 8):

1. First, it will check whether the user inputs given are blank.
2. If they are, then nothing will be saved and the user will be notified.
3. If both the fields are filled, it will allow the user to save the numbers into the database.

When the back button is pressed from Screen2, it will close the current screen and will take the user to Screen1.

You are now done with the blocks editor. The next step is to download and install the app on your phone to check how it works.

Packaging and testing

To test the app, you need to get it on your phone. First, download the application to your computer and then move it to your phone via Bluetooth or a USB cable. I'll